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1 /Users/mehakagrawal/PycharmProjects/
  Final_Dissertation/.venv/bin/python /Users/
  mehakagrawal/Desktop/Final_Dissertation/Codes/5.
  cross_dataset_direction_check/
  cross_dataset_direction_check.py
2 Cross-Dataset Direction Check: 18mo DEGs in 3mo
  data
3
4 18-month DEGs loaded: 5541 total
5   Down in MUT (18mo): 4252
6   Up in MUT (18mo): 1289
7
8 3-month genes loaded: 18522
9
10 Shared gene IDs (all): 16873
11 18mo down-in-MUT genes found in 3mo data: 2801
12 18mo up-in-MUT genes found in 3mo data: 1128
13
14 3mo log2FC values available:
15   Background (all shared genes): 16873
16   18mo down-in-MUT set: 2801
17   18mo up-in-MUT set: 1128
18
19 Statistical tests
20 18mo DOWN genes at 3mo:
21   Mean log2FC = +0.9358 (0 = no early drift)
22   One-sample t-test vs 0: t=54.232, p=0.0000
23   Mann-Whitney vs background: p=0.0000
24
25 18mo UP genes at 3mo:
26   Mean log2FC = -0.1954 (0 = no early drift)
27   One-sample t-test vs 0: t=-8.889, p=0.0000
28   Mann-Whitney vs background: p=0.0000
29
30 Saved: cross_dataset_density.png
31 Saved: cross_dataset_boxplot.png
32 Saved: cross_dataset_scatter.png
33 Saved: cross_dataset_cdf.png
34 Saved: cross_dataset_direction_summary.csv
35
36
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37 INTERPRETATION

38

39 Key question: Do the 18-month DEGs show early
directional signal at 3 months?

40

41 18mo DOWN-in-MUT genes at 3mo:

42 Mean log2FC = +0.9358

43 -> If close to 0: no early drift (age-dependent
onset confirmed)

44 -> If negative: subtle early suppression
already present

45

46 18mo UP-in-MUT genes at 3mo:

47 Mean log2FC = -0.1954

48 -> If close to 0: no early drift (age-dependent
onset confirmed)

49 -> If positive: subtle early induction already
present

50

51 One-sample t-test (vs 0):

52 Down genes: p = 0.0000 early drift detected

53 Up genes: p = 0.0000 early drift detected

54

55 Process finished successfully!

56

57 Process finished with exit code 0

58